THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** 3 x 10^5	2 x 10^30 / 6 x 10^24.
6	**(a)** 50 per cent	30/60.
7	**(d)** 2 : 1	75/25 = 3; 60/40 = 1.5; ratio 2 : 1.
8	**(a)** S^{2-}	Z = 16 is sulphur, 2 extra electrons.
9	**(b)** the theory is strongly supported but remains open to tests	Science is provisional.
10	**(a)** only one component sublimes	Different behaviour on heating.
11	**(a)** 4.4 g	Mass of gas escaped.
12	**(c)** 24.3 u	(1896 + 250 + 286)/100 = 24.32.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Some readings are slightly too high and some too low, so the mean reduces the effect of random errors.
14	**(a)** Both A and R are true and R is the correct explanation of A.	Sand stays on the filter paper while the salt solution passes through; the difference in solubility explains the method.
15	**(a)** Both A and R are true and R is the correct explanation of A.	Since the number and kind of atoms stay the same, the total mass is conserved.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Area = 96 square cm (1); About 2 per cent (1).	1+1
17	Mass of solution = 240 g (1); 24/240 x 100 = 10 per cent (1).	1+1
18	4 g H needs 32 g O (1); Water = 36 g; 8 g of oxygen left (1).	1+1
19	63x + 65(100 - x) = 6350 (1); x = 75, so the heavier isotope is 25 per cent (1).	1+1
20	A straight line shows proportionality (1); Easy to read the slope and spot errors (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Mean = 42.6/3 (1); = 14.2 cm (1); Greatest deviation = 0.1 cm (1).	1+1+1
22	320 g contains 120 g salt and 200 g water (1); At 20 degrees Celsius 200 g water holds 50 g (1); Crystals = 120 - 50 = 70 g (1).	1+1+1
23	S: 1/32 = 0.03125 (1); O: 1.6/16 = 0.1 (1); Ratio about 1 : 3: SO_3 (1).	1+1+1
24	2, 8, 3 (1); M^{3+} (1); Mass = 54 + 48 = 102 u (1).	1+1+1
	OR: Shows how reliable the result is (1); e.g. 12.3 +/- 0.1 cm (1); Allows comparison with other results (1).	1+1+1
25	A: 5/55 = 9.1 per cent (1); B: 8/108 = 7.4 per cent (1); A is more concentrated (1).	1+1+1
	OR: Mg2+ and O2- attract more strongly (1); More energy to break the lattice (1); Hence higher melting point (about 2800 vs 801 degrees Celsius) (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) 1 mm = 0.1 cm (1). (ii) $(14.2 + 14.4 + 14.3 + 14.1 + 14.5)/5 = 71.5/5 = 14.3$ cm (1). (iii) $14.3\ \text{cm} = 0.143$ m (1). (iv) Readings that are too high and too low cancel partly in the mean, so random errors are reduced (1); the largest deviation is $\lvert 14.5 - 14.3\rvert = \lvert 14.1 - 14.3\rvert = 0.2$ cm (1).	1+1+1+2
27	Run 1: O = 1.6 g, Mg : O = 3 : 2 (1); Run 2: O = 4.0 g, Mg : O = 3 : 2 (1); Same ratio in both runs (1); Formula MgO: 24 : 16 = 3 : 2 (1); Law and formula verified (1).	1+1+1+1+1
	OR: 69x + 71(100 - x) = 6980 (1); x = 60, so 60 per cent and 40 per cent (1); Neutrons: 38 and 40 (1); Element is gallium (1); Average mass between 69 and 71 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** 5.4 cm	Subtract the zero error.
3	**(b)** 32 g	Water 100 g holds 32 g; crystals 64 - 32.
4	**(b)** 3 : 8, CO_2	Atoms ratio 2.4/12 : 6.4/16 = 1 : 2.
5	**(b)** 20 and 16	X lost 2, Y gained 2.
6	**(a)** dosage of a medicine for a child	Safety critical.
7	**(a)** iron and sulphur	Only iron is magnetic.
8	**(b)** 78.5	52.5/47.5 x 71 = 78.5.
9	**(a)** 20	10x + 11(100 - x) = 1080.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Alpha particles pass through the empty space and are deflected only when they come close to the tiny nucleus.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the SI unit does not explain why the least count is 1 mm (it is the smallest division on the scale).
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true statements, but R does not explain why a fractionating column is needed in A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Boiling point is fixed (1); Heat supplies latent heat (1).	1+1
17	Na+: 2, 8; F-: 2, 8 (1); Same as neon (1).	1+1
18	Maximum 2n squared with n = 1 is 2 (1); K shell is the smallest and nearest to the nucleus (1).	1+1
19	Distance = 3 x 10^5 x 3.15 x 10^7 (1); = 9.45 x 10^12 km (1).	1+1
20	Slow cooling allows ordered growth (1); Few nuclei form (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	p + n = 31 and n = p + 5 (1); 2p = 26, so p = 13 and n = 18 (1); Z = 13; configuration 2, 8, 3 (1).	1+1+1
22	Rising curve (1); Not a straight line (1); Longer pendulum has longer period (1).	1+1+1
23	Salt gives ions that neutralise the charge on particles (1); Particles aggregate (1); Soap separates (1).	1+1+1
24	Molecular mass = 12 + 4 = 16 u (1); Carbon = 12/16 x 100 = 75 per cent (1); Hydrogen = 4/16 x 100 = 25 per cent (1).	1+1+1
	OR: Electrons occupy fixed shells (1); No radiation in these states (1); Energy changes only by jumps (1).	1+1+1
25	Mean = 54.4 s for 30 (1); T = 1.81 s (1); Spread about 0.4 / 54.4 = 0.7 per cent (1).	1+1+1
	OR: X = 0.27 (1); Y = 0.6, Z = 0.8 (1); X is most strongly adsorbed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

26	(i) Solute: salt; solvent: water (1). (ii) $20 + 180 = 200$ g (1). (iii) $\frac{20}{200} \times 100 = 10\%$ (1). (iv) For 5 per cent, total mass $= \frac{20}{0.05} = 400$ g (1); water to be added $= 400 - 200 = 200$ g (1).	1+1+1+2
27	Atoms of different elements have different masses (1); 12 g of Mg has half as many atoms as 12 g of C (1); Number of atoms is proportional to mass divided by atomic mass (1); 24 g of Mg has as many atoms as 12 g of C (1); Atomic masses link mass to number of atoms (1).	1+1+1+1+1
	OR: n = 24 - 12 = 12 (1); B: p = 11, n = 12, A = 23 (1); B is sodium $^{23}_{11}\mathrm{Na}$ (1); 2, 8, 1 (1); A is magnesium-24 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 28 g	28/44 of the mass is nitrogen.
2	**(d)** argon	18 electrons.
3	**(a)** check the experiment and the assumptions in the model	Investigate.
4	**(b)** dissolve in water, filter, evaporate	Sugar dissolves.
5	**(c)** 27.3 per cent	12/44 x 100 = 27.3.
6	**(c)** electron	An electron is about 1/1836 of the mass of a proton.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(c)** 15 per cent	3 x 5 per cent.
12	**(b)** the one with R_f 0.9	Largest R_f.
13	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the fixed ratio is explained by the law of constant proportions, not just by water being a compound.
14	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the discovery of the neutron is not the reason for the model's failure (a revolving electron should lose energy and spiral into the nucleus).
15	**(c)** A is true but R is false.	Models keep only the features that matter for the question, so they are not exact copies; R is false.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	117/58.5 = 2 (1); Twice as many (1).	1+1
17	No dense positive centre in his model (1); It predicted only small deflections (1).	1+1
18	Mean = 9.76 m/s^2 (1); Range = 0.3 m/s^2 (1).	1+1
19	Solute may have low solubility (1); e.g. a saturated solution of chalk or CO2 in water is dilute (1).	1+1
20	Formula unit mass of CuO = 64 + 16 = 80 u (1); So 64 g of copper is present in 80 g (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Mass of O2 = 24.5 - 14.9 = 9.6 g (1); Percentage = 9.6/24.5 x 100 (1); = 39.2 per cent (1).	1+1+1
22	Same valence electrons in a group (1); Across a period valence electrons increase 1 to 8 (1); Chemical behaviour depends on valence electrons (1).	1+1+1
23	Volume = 144 cubic m (1); Mass = 144 x 1.2 (1); = about 173 kg (1).	1+1+1
24	Ink spots from document and suspects (1); Run chromatography (1); Compare bands and R_f values (1).	1+1+1
	OR: Weighing error (1); Leakage of a gas (1); The law holds when all products are weighed (1).	1+1+1
25	Splits on neutron capture giving energy (1); Chain reaction controlled in reactors (1); Radioactive waste and accidents (1).	1+1+1
	OR: Informed consent (1); Safety tests first (1); Honest reporting of side effects (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) In a chemical reaction the total mass of the reactants equals the total mass of the products (mass is neither created nor destroyed) (1). (ii) $10 - 5.6 = 4.4$ g (1). (iii) Antoine Lavoisier (1). (iv) No (1); carbon dioxide escapes into the air from the open dish, and if its mass is included the total mass stays 10 g (1).	1+1+1+2
27	Conservation: atoms rearranged (1); Constant proportions: fixed ratios of atoms (1); Atoms are divisible (1); Isotopes differ in mass (1); Atoms combine in whole-number ratios remains valid (1).	1+1+1+1+1
	OR: Closest approach of alpha particles gives an upper limit for the nucleus size (1); Atom radius about 10^-10 m (1); Density = mass/volume is huge (1); About 10^18 kg per cubic metre (1); Nucleus holds nearly all the mass (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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